Reserve-a-Lab System Evaluation Form (ISO/IEC 25010:2023)

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How to use this form

This form rates Reserve-a-Lab on the nine product-quality characteristics of ISO/IEC 25010:2023, using 69 items on a 5-point Likert scale. Every rating in every item has a fixed, measurable criterion, so two evaluators testing the same build should arrive at the same score.

Evaluator information

Field	Entry
Name	
Evaluator type (IT expert / Admin / Teacher / Requester)	
Date of evaluation	
System version or commit tested	
Device, OS and browser used	
Network used (school LAN / Wi-Fi / mobile data) and measured speed	

Rating scale

Score	Rating	General meaning
5	Excellent	Meets or beats the target in every trial; nothing to fix.
4	Very Good	Meets the target with minor, cosmetic shortfalls.
3	Good	Acceptable for daily use; noticeable gaps that users can work around.
2	Fair	Frequent problems; users need help or workarounds often.

Score	Rating	General meaning
1	Poor	Fails the criterion or the feature is missing.

Who answers which item

- **[U]** items can be rated by end users (requesters, teachers, admins) after using the system.
- **[T]** items need a technical evaluator (IT faculty, developer, or IT professional) who can use browser DevTools, read the code or inspect the database.
- An evaluator who cannot test an item writes **N/A** and that item is left out of the mean.

Test conditions

1. Test the deployed system, not a developer's local copy, unless the local copy is the only version.
2. Use a laptop with Google Chrome (latest) for timed items, and one phone (≤ 430 px wide) for responsiveness items.
3. For every timed item, run **10 trials** and record the **median** (middle value); clear the browser cache before the first trial only.
4. Measure time with Chrome DevTools → Network tab ("Finish" or the XHR's "Time") or a stopwatch where noted.
5. Seed the database with realistic data before testing: at least 10 labs, 300 requests and 200 inventory items.

1. Functional Suitability

ISO/IEC 25010:2023 defines functional suitability as the capability of a product to provide functions that meet stated and implied needs of intended users (3.1), split into completeness, correctness and appropriateness.

#	Item and how to test	5	4	3	2	1	Score
FS1	**Functional completeness [T/U]** — Tick each function in the checklist below that exists and works end to end. Score = ticked ÷ total.	100%	95–99%	85–94%	70–84%	Below 70%	
FS2	**Functional correctness: general [T]** — Run 30 prepared test cases (at least 10 per role) with known expected results. Score = % that give the expected result.	100%	97–99%	90–96%	80–89%	Below 80%	
FS3	**Functional correctness: booking conflicts [T]** — Submit 10 requests that overlap an approved or pending booking (same lab, same date, overlapping time, incl. 1-minute overlaps), plus 2 back-to-back requests that should be allowed.	All 10 flagged and both back-to-back allowed	9 of 10 flagged	8 of 10 flagged	6–7 flagged	5 or fewer flagged	
FS4	**Functional correctness: inventory quantities [T]** — Perform 20 stock-ins, edits and CSV stock-ins on 10 items. Compare the quantity shown with the hand-computed quantity.	0 mismatches	1 mismatch	2 mismatches	3–4 mismatches	5 or more	

#	Item and how to test	5	4	3	2	1	Score
FS5	**Functional correctness: reports and printed forms [T]** — Generate the summary PDF for one month and print 5 request forms. Compare every count, name, date and control number with the database.	All values match	Formatting slip only, no wrong value	1 wrong value	2–3 wrong values	4 or more, or report fails	
FS6	**Functional appropriateness: approval path [U]** — Count clicks from the dashboard to a completed approval (open request → review → approve with feedback → confirm), for teacher and admin. Use the higher count.	4 clicks or fewer	5–6	7–8	9–10	More than 10	
FS7	**Functional appropriateness: request form [U]** — Count the fields on the request form that the system already knows but still makes the requester type (name, ID, email, grade, section, school year, today's date).	0	1	2	3	4 or more	

FS1 checklist: required functions

Requester

- [ ] Log in and log out

- [ ] View laboratories with photos, location and availability
- [ ] View the reservation calendar
- [ ] Submit a request with date, time, subject, students and materials
- [ ] See a conflict warning before submitting
- [ ] Track the approval status of each request
- [ ] Print the approved request form
- [ ] Receive email on submission and on each decision
- [ ] Change password and reset a forgotten password

Teacher

- [ ] View requests assigned to them
- [ ] Approve or reject with feedback
- [ ] Receive email when a request is assigned

Admin

- [ ] Add, edit, deactivate a laboratory and manage its gallery
- [ ] Set laboratory availability
- [ ] Approve, force-approve or reject any request, with control numbers
- [ ] Add, edit, remove inventory items
- [ ] Stock in by form, by CSV and by barcode scan
- [ ] Set alert thresholds and receive low-stock alerts
- [ ] Generate the summary report as PDF
- [ ] Set the current school year

Total: 20 functions. Add any function your requirements document lists that is missing here.

2. Performance Efficiency

Performance efficiency is the capability to perform functions within specified time and throughput parameters while using resources efficiently (3.2), split into time behaviour, resource utilization and capacity. Time limits follow Nielsen's 0.1 s / 1 s / 10 s response limits and Google's Core Web Vitals (LCP good ⩽ 2.5 s, poor > 4 s; INP good ⩽ 200 ms, poor > 500 ms).

#	Item and how to test	5	4	3	2	1	Score
PE1	**Time behaviour: login [U/T]** — Time from clicking Log in to the dashboard being fully shown. Median of 10 trials.	1.0 s or less	1.01–1.5 s	1.51–2.5 s	2.51–4.0 s	More than 4.0 s	
PE2	**Time behaviour: page load [T]** — Largest Contentful Paint (Chrome DevTools → Lighthouse, desktop) for home, request form, request tracker, calendar, approval page and inventory. Use the average of the 6 pages.	1.5 s or less	1.51–2.5 s	2.51–3.0 s	3.01–4.0 s	More than 4.0 s	
PE3	**Time behaviour: in-page actions [T]** — Server response time (Network tab, XHR "Time") for: load calendar events, conflict check, approve, reject, stock in, open request details. Median of all.	200 ms or less	201–500 ms	501 ms–1.0 s	1.01–2.0 s	More than 2.0 s	
PE4	**Time behaviour: request submission [U/T]** — Time from clicking Submit on a full request (5 materials, 20 students) to the confirmation message.	2.0 s or less	2.01–3.0 s	3.01–5.0 s	5.01–10.0 s	More than 10 s	
PE5	**Time behaviour: email notification [U]** — Time from submitting or approving a request to the email appearing in the recipient's Gmail inbox. Median of 5 trials.	30 s or less	31–60 s	1–3 min	3–10 min	More than 10 min, or not received	
PE6	**Time behaviour: PDF generation [U/T]** — Time from clicking Print/Generate to the PDF opening, for one request form and one monthly summary report. Use the slower of the two.	3 s or less	3.01–5 s	5.01–8 s	8.01–15 s	More than 15 s	

#	Item and how to test	5	4	3	2	1	Score
PE7	**Time behaviour: search and filter [U]** — Time for the request and inventory tables to update after typing in the search box, with 300 requests and 200 items loaded.	Instant (0.1 s or less)	0.11–0.3 s	0.31–1.0 s	1.01–2.0 s	More than 2.0 s	
PE8	**Resource utilization: page weight [T]** — Data transferred on first, uncached load (Network tab footer), average of the 6 pages in PE2.	1 MB or less	1.01–2 MB	2.01–3 MB	3.01–5 MB	More than 5 MB	
PE9	**Capacity: concurrent users [T]** — Load test (Apache JMeter, k6 or ApacheBench) on login + home + request tracker. Record the highest number of simultaneous users at which 95% of responses stay under 2 s and errors stay under 1%.	100 users or more	50–99	30–49	15–29	Fewer than 15	
PE10	**Capacity: data volume [T]** — Seed 5,000 requests and 1,000 inventory items, then load the admin request tracker and inventory page. Use the slower page.	2.5 s or less	2.51–3.0 s	3.01–4.0 s	4.01–6.0 s	More than 6 s, or page fails	

Why these numbers: 1 s is the limit for the user's flow of thought to stay unbroken, so a login under 1 s earns full marks. 4 s is Google's "poor" line for page load, so anything slower scores 1. 10 s is the limit of user attention, so a submission over 10 s scores 1.

3. Compatibility

Compatibility is the capability to exchange information with other products and to work while sharing a common environment and resources (3.3), split into co-existence and interoperability.

#	Item and how to test	5	4	3	2	1	Score
C1	**Co-existence: shared server [T]** — Deploy the system on the same Apache/MySQL server as one other PHP application. Check for clashes in port, database, session cookie name, and folder paths.	No clash; own database, own DB user, own session name	No clash; shares a default session name or DB user	1 clash fixed by configuration only	1 clash that needs a code change	Cannot run beside another app	
C2	**Interoperability: email clients [U]** — Open the submission, approval and low-stock emails in Gmail (web), Gmail (Android/iOS app) and Outlook.	Correct layout and working links in all 3	All 3 readable, minor layout slips	2 of 3 correct	1 of 3 correct	None correct, or emails land in spam	
C3	**Interoperability: CSV import [T]** — Import a 100-row inventory CSV saved from Excel, including 5 deliberately bad rows and values with commas inside quotes.	All good rows imported; bad rows listed in an error report	All good rows imported; no error report	95–99% of good rows imported	80–94% imported	Below 80%, or import fails	
C4	**Interoperability: export [U]** — Export reports and request lists. Open each file in Chrome/Acrobat and in Excel or Google Sheets.	PDF plus CSV or Excel, all open without error	PDF plus one other format, minor issues	PDF only, correct	PDF with layout or data errors	No export	

#	Item and how to test	5	4	3	2	1	Score
C5	**Interoperability: barcode scanner [U]** — Scan 20 inventory items (EAN-13 and Code 128 labels) with a USB scanner or phone camera.	20 of 20 read on first scan	19 of 20	17–18	14–16	13 or fewer, or not supported	
C6	**Interoperability: school records [T]** — Compare 20 samples of student, section and subject data pulled into the request form with the source tables.	20 of 20 match	19 of 20	18 of 20	16–17	15 or fewer	

4. Interaction Capability

Interaction capability replaces "usability" in the 2023 edition: the capability of a product to be interacted with by specified users to complete the intended task (3.4). It has eight subcharacteristics; each has at least one item below. User-based items need **at least 5 first-time users per role**; record the average.

#	Item and how to test	5	4	3	2	1	Score
IC1	**Appropriateness recognizability [U]** — Show 5 first-time users the login and home pages for 30 s. Ask each to name 3 things the system does.	5 of 5 name 3 correct functions	4 of 5	3 of 5	2 of 5	1 or 0 of 5	

#	Item and how to test	5	4	3	2	1	Score
IC2	**Learnability: time to first request [U]** — A first-time requester with no training submits a complete request (lab, date, time, subject, 3 materials, 10 students). Average time.	3 min or less	3–5 min	5–8 min	8–12 min	More than 12 min, or needs help	
IC3	**Learnability: unaided task success [U]** — Give each first-time user 5 core tasks for their role. Score = % of tasks finished correctly without asking for help.	100%	90–99%	80–89%	65–79%	Below 65%	
IC4	**Operability: SUS score [U]** — Each user answers the 10-item System Usability Scale after testing. Average the SUS scores (0–100). Average SUS across 500 studies is 68; above 80.3 is the top 10%.	80.3 or higher	74.0–80.2	68.0–73.9	51.0–67.9	Below 51	
IC5	**Operability: consistency [T]** — Check 10 common controls across all pages (Save, Cancel, Approve, Reject, Delete, Close, Search, Back, Print, Logout) for the same label, color and position. Count inconsistencies.	0	1–2	3–4	5–6	7 or more	

#	Item and how to test	5	4	3	2	1	Score
IC6	**User error protection: input validation [T]** — Enter 20 invalid inputs: past date, end time before start, negative or zero quantity, blank required field, bad email, text in number field, 10 MB image, wrong file type.	20 of 20 blocked with a clear message	18–19	15–17	10–14	Fewer than 10	
IC7	**User error protection: destructive actions [T]** — List every action that rejects, deletes, removes, deactivates or force-approves. Score = % that ask for confirmation (or allow undo).	100%	80–99%	60–79%	40–59%	Below 40%	
IC8	**User engagement: visual design checklist [T/U]** — Tick: (1) school colors used consistently, (2) at most 2 font families, (3) elements aligned to a grid, (4) consistent spacing, (5) one icon style, (6) status shown by color **and** text, (7) no broken images, (8) no overlapping elements, (9) designed empty states ("No requests yet"), (10) consistent buttons and cards.	10 of 10	8–9	6–7	4–5	3 or fewer	

#	Item and how to test	5	4	3	2	1	Score
IC9	**Inclusivity: accessibility score [T]** — Lighthouse Accessibility score, average of the 6 main pages. Text contrast must meet WCAG 2.2 AA (4.5:1 normal text, 3:1 large text).	95–100	90–94	80–89	70–79	Below 70	
IC10	**Inclusivity: keyboard-only use [T]** — Using only Tab, Shift+Tab, Enter, Space and arrow keys, submit a request and approve a request.	Both tasks done; focus outline visible on every control	Both done; focus outline missing in 1–2 places	One task done	Partly done, keyboard traps found	Neither task possible	
IC11	**User assistance [U]** — Tick: (1) user manual or help page, (2) every input has a visible label, (3) format hints on date, time and quantity fields, (4) tooltips on icon-only buttons, (5) support contact shown.	5 of 5	4 of 5	3 of 5	2 of 5	1 or 0	
IC12	**Self-descriptiveness: feedback [U]** — Perform 10 actions (submit, approve, reject, stock in, upload image, change password, etc.). Count those that show a clear success or failure message within 1 s, and requests that show their current approval stage.	10 of 10	9 of 10	7–8	5–6	4 or fewer	

#	Item and how to test	5	4	3	2	1	Score
IC13	**Self-descriptiveness: error messages [T]** — Trigger 10 errors. Count messages that say what went wrong and how to fix it, in plain words, with no raw PHP, SQL or JSON shown.	10 of 10	8–9	6–7	4–5	3 or fewer	

5. Reliability

Reliability is the capability to perform specified functions under specified conditions for a specified period, split into faultlessness, availability, fault tolerance and recoverability. The 2023 edition renamed "maturity" to faultlessness.

#	Item and how to test	5	4	3	2	1	Score
R1	**Faultlessness: walkthrough errors [T]** — Walk through every page and action for all 3 roles with the browser Console open and PHP error logging on. Count errors and warnings.	0 errors, 0 warnings	Warnings only (1–2), no errors	1–3 errors with no visible effect	4–6 errors, or any visible malfunction	More than 6, or a blank or crashed page	
R2	**Faultlessness: pilot defects [U]** — Run a 1-week pilot with real users. Classify reported defects: critical = data loss or cannot log in; major = a function fails; minor = cosmetic.	0 major, 2 or fewer minor	0 major, 3–5 minor	1 major	2–3 major	Any critical, or more than 3 major	

#	Item and how to test	5	4	3	2	1	Score
R3	**Availability: uptime [T]** — Monitor the live URL for 30 days with a 5-minute uptime checker (e.g. UptimeRobot).	99.9% or higher (⩽ 44 min down per month)	99.5–99.89% (⩽ 3 h 39 min)	99.0–99.49% (⩽ 7 h 18 min)	98.0–98.99%	Below 98%	
R4	**Fault tolerance: dependency failures [T]** — Inject 3 faults: (a) wrong SMTP password, then submit a request; (b) stop MySQL, then open a page; (c) cut the network mid-submission.	3 of 3 handled: no data lost, clear message shown	3 of 3 handled, message unclear	2 of 3 handled	1 of 3 handled	0 of 3, or raw errors shown	
R5	**Fault tolerance: duplicate and simultaneous submissions [T]** — (a) Double-click Submit; (b) two users book the same lab slot from two browsers within 1 s.	Both prevented, losing user told why	Both prevented, no explanation	One of the two prevented	Duplicates saved but flagged to admin	Duplicates saved silently	
R6	**Recoverability: backup and restore [T]** — Check the backup schedule, then restore the database from the latest backup on a test server. Time the restore.	Automatic daily backup; restore 30 min or less	Automatic weekly or manual daily backup; restore 1 h or less	Documented manual backup; restore 2 h or less	Backup exists but restore untested or over 4 h	No backup	

#	Item and how to test	5	4	3	2	1	Score
R7	**Recoverability: interrupted form [U]** — Fill half of the request form, then refresh the page, or let the session expire.	All entries kept, user returns to the same step	Most entries kept	Warning shown before data is lost (e.g. timeout warning)	Data lost, clean redirect to login	Data lost with an error page	

6. Security

Security is the capability to protect information and data from attack and to give each person only the access their role allows. It has six subcharacteristics; the 2023 edition added **resistance** (keeps working while under attack). All items are [T]; test only on a copy of the system, never on live data.

#	Item and how to test	5	4	3	2	1	Score
S1	**Confidentiality: role-based access** — Try 15 restricted pages and AJAX actions as the wrong role and while logged out (e.g. requester opens `admin_home.php`, `inventory.php`, or posts `approve` to the admin action endpoint).	15 of 15 denied	14 denied; the leak is read-only, non-personal data	14 denied	12–13 denied	11 or fewer denied	
S2	**Confidentiality: password storage** — Inspect the `accounts` table and login code. OWASP recommends Argon2id or bcrypt; MD5 is unsuitable.	Argon2id or bcrypt via `password_hash()`	bcrypt or PBKDF2 with weak settings	Salted SHA-256	Unsalted SHA-family hash	MD5 or plain text	

#	Item and how to test	5	4	3	2	1	Score
S3	**Confidentiality: transport and cookies** — Tick: (1) HTTPS with valid certificate, (2) HTTP redirects to HTTPS, (3) HSTS header, (4) session cookie `HttpOnly`, (5) session cookie `Secure` and `SameSite`.	5 of 5	4 of 5	3 of 5	2 of 5	1 or 0	
S4	**Integrity: injection** — Enter 20 SQL-injection and XSS payloads (e.g. `' OR 1=1 --`, `<script>alert(1)</script>`) into every text field, URL parameter and CSV.	0 succeed; all queries are prepared statements	0 succeed; a few queries built by string joining	1 XSS succeeds; no SQL injection	2+ XSS, or SQL injection that reveals no data	SQL injection that reveals data or bypasses login	
S5	**Integrity: CSRF protection** — List every endpoint that changes data (submit, approve, reject, stock in, add/edit lab, delete, change password). Score = % that reject requests without a valid CSRF token.	100%	90–99%	70–89%	40–69%	Below 40%	
S6	**Integrity: file uploads** — Upload (a) a `.php` file renamed `.jpg`, (b) a 20 MB image, (c) an SVG with embedded script, as a lab image.	All 3 rejected; files stored under random names in a non-executable folder	All 3 rejected	2 of 3 rejected	1 of 3 rejected	None rejected, or the PHP file runs	

#	Item and how to test	5	4	3	2	1	Score
S7	**Non-repudiation and accountability: audit trail** — Check that 6 action types record the user ID and timestamp: submit, approve, reject, force-approve, inventory change, lab edit.	6 of 6 recorded and viewable by admin	6 of 6 recorded in database only	4–5 recorded	2–3 recorded	1 or 0	
S8	**Authenticity: login and session** — Tick: (1) same error for wrong email and wrong password, (2) lockout or delay after 5 failed attempts, (3) session ID regenerated at login, (4) idle timeout of 30 min or less, (5) reset links expire within 60 min and work once.	5 of 5	4 of 5	3 of 5	2 of 5	1 or 0	
S9	**Resistance: automated scan** — Run the OWASP ZAP baseline scan against the test copy. Count alerts by risk level.	0 High, 0 Medium	0 High, 1–3 Medium	0 High, 4+ Medium	1 High	2 or more High	

7. Maintainability

Maintainability is how effectively and efficiently the product can be changed to fix, improve or adapt it, split into modularity, reusability, analysability, modifiability and testability. All items are [T] and exclude third-party libraries (Bootstrap, jQuery, DataTables, PHPMailer, Dompdf, `vendor/`).

#	Item and how to test	5	4	3	2	1	Score
M1	**Modularity: shared components** — Score = % of pages that load the header, footer, database connection and session check from shared include files instead of copying them.	100%	90–99%	75–89%	50–74%	Below 50%	
M2	**Modularity: file size** — Score = % of the project's own PHP and JS files with 500 lines or fewer (count with `wc -l` or an IDE).	90% or more	80–89%	65–79%	50–64%	Below 50%	
M3	**Reusability: code duplication** — Run `jscpd` (or PHPCPD) over the project's own code. Read the duplicated-lines %.	3% or less	3.1–5%	5.1–10%	10.1–20%	More than 20%	
M4	**Analysability: documentation and diagnostics** — Tick: (1) README with setup steps, (2) database schema or ERD, (3) comments on at least half of the functions, (4) errors written to a log and hidden from users, (5) Git history with meaningful messages.	5 of 5	4 of 5	3 of 5	2 of 5	1 or 0	

#	Item and how to test	5	4	3	2	1	Score
M5	**Modifiability: standard change** — A developer new to the code adds a field (e.g. "Purpose of activity") to the request form, saves it, and shows it in the tracker and printout. Time it, then re-run FS2.	1 h or less, nothing broken	1–2 h, nothing broken	2–4 h, nothing broken	4–8 h, or 1 regression	More than 8 h, or 2+ regressions	
M6	**Modifiability: central configuration** — Count database credentials, SMTP settings, URLs and school-year values hard-coded outside the config files.	0	1–2	3–5	6–10	More than 10	
M7	**Testability: test coverage** — Check for tests of the 5 critical functions: login, submit, conflict check, approve/reject, inventory stock-in.	Automated tests cover all 5 and run on each push (CI)	Automated tests cover 3–4	Written manual test cases for all 5	Test cases for 1–4	No test cases	

8. Flexibility

Flexibility replaces "portability" in the 2023 edition: the capability to adapt to changes in requirements, contexts of use or system environment. It adds **scalability** to adaptability, installability and replaceability.

#	Item and how to test	5	4	3	2	1	Score
F1	**Adaptability: browsers [U/T]** — Do the 5 core tasks on Chrome, Edge and Firefox (Windows), Safari (iPhone) and Chrome (Android).	All 5 browsers, no issues	All 5, cosmetic issues only	4 of 5 browsers	3 of 5	2 or fewer	
F2	**Adaptability: screen sizes [U/T]** — Open each of 8 main pages at 375 px (phone), 768 px (tablet) and 1366 px (laptop): 24 checks. A check passes with no sideways scrolling, no cut-off controls and the main content visible.	24 of 24	22–23	20–21	16–19	15 or fewer	
F3	**Scalability: growth without code changes [T]** — Through the admin screens only, try to: add a laboratory, start a new school year, add a subject or section, add an inventory category, add a user account.	5 of 5	4 of 5	3 of 5	2 of 5	1 or 0	
F4	**Installability [T]** — An IT staff member who did not build the system installs it on a fresh XAMPP or server using only the README: copy files, import database, set config, first login.	30 min or less, no help	31–60 min, no help	1–2 h, no help	2–4 h, or needs the developer's help	Cannot install	

#	Item and how to test	5	4	3	2	1	Score
F5	**Replaceability: paper process [U]** — Compare the system with the current paper lab request and permit forms. Score = % of the paper fields and signatures (incl. SDS check and control numbers) the system captures.	100%, and prints in the official format	100%, different print format	90–99%	75–89%	Below 75%	

9. Safety

Safety is new in the 2023 edition: the capability to avoid states that endanger human life, health, property or the environment. For a science-lab system the hazards are unsupervised lab use, double-booked rooms and unchecked chemicals, so the items below test those.

#	Item and how to test	5	4	3	2	1	Score
SF1	**Operational constraint [T]** — Try to book: (a) a lab marked unavailable or inactive, (b) an already-approved slot, (c) outside school hours, (d) without a supervising teacher.	4 of 4 blocked	3 of 4	2 of 4	1 of 4	None blocked	

#	Item and how to test	5	4	3	2	1	Score
SF2	**Risk identification: hazardous materials [U/T]** — Request a hazardous reagent (e.g. hydrochloric acid). Check how the system identifies it before approval.	Auto-flagged from inventory; approver must confirm the SDS before approving	Flagged on screen; SDS confirmed on the printed form	SDS column on the printed form only	Materials recorded as free text; no flag	Materials not recorded	
SF3	**Hazard warning: low stock and expiry [U/T]** — Set a threshold, then stock out below it. Note when the admin is warned.	Dashboard alert at once and email within 5 min	Dashboard and email within 1 h	One channel only (dashboard or email)	Warning only when admin opens inventory	No warning	
SF4	**Fail safe [T]** — Inject 5 failures during approval and stock-in: session expires, DB error, email error, invalid request ID, interrupted request. Check the record afterwards.	5 of 5 stay in the last valid state (never falsely approved, no partial update)	4 of 5	3 of 5	2 of 5	1 or 0, or a request ends up falsely approved	
SF5	**Safe integration [T]** — Block internet access (CDNs) and the SMTP server. Then submit, run a conflict check, approve, and stock in.	4 of 4 work, with notice that email is queued or failed	4 of 4 work, no notice	3 of 4	2 of 4	1 or 0	

Scoring summary

Each characteristic's score is the mean of its rated items; the overall score is the mean of the nine characteristic means, so every characteristic counts equally no matter how many items it has. Items marked N/A are left out.

$$\bar{x}_{c} = \frac{\sum_{i=1}^{n} s_i}{n} \qquad \bar{x}_{overall} = \frac{\sum_{c=1}^{9} \bar{x}_{c}}{9}$$

Here s is an item score, n the number of rated items in a characteristic, and c a characteristic. With several evaluators, average each item across evaluators first.

Characteristic	Items	Items rated	Mean	Interpretation
1. Functional Suitability	FS1–FS7 (7)			
2. Performance Efficiency	PE1–PE10 (10)			
3. Compatibility	C1–C6 (6)			
4. Interaction Capability	IC1–IC13 (13)			
5. Reliability	R1–R7 (7)			
6. Security	S1–S9 (9)			
7. Maintainability	M1–M7 (7)			
8. Flexibility	F1–F5 (5)			
9. Safety	SF1–SF5 (5)			
Overall	**69 items**			

Verbal interpretation

Mean range	Interpretation	Meaning
4.21–5.00	Excellent	Ready for deployment as is.
3.41–4.20	Very Good	Ready for deployment; minor fixes recommended.
2.61–3.40	Good	Usable; fix the lowest-scoring items before full rollout.
1.81–2.60	Fair	Needs major revision before deployment.
1.00–1.80	Poor	Not acceptable; rework required.

The ranges divide the 1–5 scale into five equal steps of 0.80. Any Security (S1–S9) or Safety (SF1–SF5) item scored 1 should be fixed before deployment, whatever the mean.

Comments and recommendations

Evaluator certification

I certify that I tested the system under the conditions above and that these scores reflect my measurements.

Evaluator signature over printed name	Date

Sources

- ISO/IEC 25010:2023, *Systems and software engineering — SQuaRE — Product quality model* (uploaded preview, clauses 1–3.4), and ISO's catalogue page
- ISO 25000 portal: ISO/IEC 25010 — definitions of the remaining characteristics and subcharacteristics
- Nielsen Norman Group, Response Times: The 3 Important Limits — 0.1 s, 1 s, 10 s
- web.dev, How the Core Web Vitals metrics thresholds were defined — LCP, INP, CLS
- MeasuringU, Measuring Usability with the System Usability Scale — average SUS 68, top 10% above 80.3, bottom 15% below 51
- W3C, Understanding WCAG 2.2: Contrast (Minimum) — 4.5:1 and 3:1

- Chrome for Developers, Lighthouse accessibility scoring
- OWASP, Password Storage Cheat Sheet — Argon2id, bcrypt; MD5 unsuitable
- uptime.is: 99.5% SLA — downtime allowed per month